FlowSense — Requirements Summary

Source: Distilled from the email correspondence with Clear The Air.

1. Core Product

- Non-intrusive device clamped externally to pipework.
- Primarily designed for 15 mm copper pipe.
- 28 mm pipe also needs consideration for higher flow ranges.
- Ideally, the same device should monitor both hot and cold pipes when they are adjacent.
- The design must be cost-effective enough for customers to deploy at scale.

2. Temperature

- Target accuracy: within 1%.
- Manufacturer calibration certification would be ideal.
- Temperature data is to be recorded in 15-minute intervals.

3. Flow / Event Detection

- **240 V version:** transmit data packets when Tap ON events are detected.
- **240 V version:** upload historical data every 2 hours, containing 15-minute records for the time graph.
- **Battery version:** upload normal 15-minute readings once per day.
- **Battery version:** transmit Tap events as required.

4. Connectivity

- NB-IoT is required/preferred.
- BLE and Wi-Fi are considered unsuitable because they are not sufficiently stable or secure for this application.
- Clear The Air normally uses 1NCE eSIMs.
- Ideally, NB-IoT/4G connectivity should be integrated into one device rather than using a separate communications unit.

5. Power

- Preferred solution: battery powered.
- Minimum target battery life: 3 years.
- 5–10 years would be desirable if achievable.
- 240 V is an alternative where available, but mains power may increase installation and customer costs.

6. Data & Platform

- Clear The Air already has its own PC platform, apps and servers.
- The FlowSense device will send data to Clear The Air's existing servers.
- Clear The Air will handle data interpretation and presentation.

- There is therefore no current requirement for FlowSense to develop its own app or cloud platform.

7. Commercial Target

- Clear The Air currently pays approximately £80–£135 per NB-IoT telematics device.
- The FlowSense device needs to be cost-effective relative to this range.
- Clear The Air has specifically requested the best estimate of unit pricing.

8. Development Stages

- **Stage 1 — current:** develop a working flow/temperature prototype, targeted for completion around the end of September.
- **Stage 1 review:** demonstrate testing, identify issues and discuss improvements, potentially using a short video.
- **Stage 2:** add NB-IoT communications and integrate the device with Clear The Air's server infrastructure.
- Stage 2 will also finalise the data-transmission and power-management strategy.

Overall Product Definition

A low-cost, battery-powered, non-invasive 15 mm/28 mm pipe device that measures temperature to within 1%, detects tap-flow events, records 15-minute data, and transmits it via NB-IoT/eSIM to Clear The Air's existing platform, with a minimum 3-year battery life.

Key Engineering Challenges

- Achieving NB-IoT connectivity while maintaining a 3-year minimum battery life.
- Achieving the required temperature accuracy in a non-invasive, externally mounted pipe design.
- Keeping the unit cost sufficiently low for large-scale deployment.
- Supporting both 15 mm and 28 mm pipe installations, potentially with hot and cold measurement on one device.